

Telemetry Protocol Specification

The AirOne wire format is a compact binary frame with a fixed header, a variable payload, an optional 8-byte authentication tag, and a trailing CRC32. It is implemented in `src/telemetry/protocol.py`.

Frame layout

Offset Size Field

Offset	Size	Field
0	4	MAGIC = A1 60 4E 45
4	1	VERSION = 0x71
5	1	PACKET_TYPE
6	4	SEQUENCE_NUMBER (uint32, little-endian)
10	8	TIMESTAMP (uint64, little-endian)
18	2	PAYLOAD_LENGTH N (uint16, little-endian)
20	1	FLAGS (bit0=FEC_PRESENT, bit1=COMPRESSED, bit2=ENCRYPTED, bit3=AUTHENTICATED)
21	N	PAYLOAD
21+N	8	AUTH_TAG — only when FLAGS bit3 is set
21+N+T	4	CRC32 over every preceding byte (header, payload, tag) (uint32, LE)

PAYLOAD_LENGTH counts the payload only (never the tag). Unknown VERSION, PACKET_TYPE or FLAGS bits cause the frame to be rejected and counted (`invalid_version` / `invalid_type` / `invalid_flags`) — never parsed.

- Header size: **21 bytes**. CRC size: **4 bytes**.
- Header struct format: `<4sBBIQHB`.
- Maximum payload: **65535 bytes** (bounded by the uint16 length field).

Constants

Name	Value
MAGIC	A1 60 4E 45
VERSION	0x71
HEADER_SIZE	21
CRC_SIZE	4
MAX_PAYLOAD	65535
FLAG_FEC_PRESENT	0x01
FLAG_COMPRESSED	0x02
FLAG_ENCRYPTED	0x04
FLAG_AUTHENTICATED	0x08
AUTH_TAG_SIZE	8
MIN_LINK_KEY_BYTES	16

Packet types (PacketType)

Value	Name
0x01	SENSOR_DATA
0x02	GPS
0x03	SYSTEM_STATUS
0x04	COMMAND
0x05	ACK
0x06	HEARTBEAT
0x07	FEC_DATA
0x08	ERROR

CRC32

The CRC32 is computed with `binascii.crc32` masked to 32 bits and packed little-endian. It covers **every byte preceding the CRC field** — the full header and the payload. A frame whose recomputed CRC does not match is marked `crc_valid = False`; downstream, the affected data is tagged `CORRUPTED` and is never treated as a real reading.

Frame authentication (FLAG_AUTHENTICATED)

`AUTH_TAG = HMAC-SHA256(link_key, header || payload)[0:8]`. The key (≥ 16 bytes, hex) is `AIRONE_LINK_KEY` on the ground station and `AIRONE_LINK_KEY_HEX` in the firmware; `compute_auth_tag()` / `airone_auth_tag()` are byte-for-byte identical (tested against RFC 4231). The tag is compared in constant time. The resulting `auth_state` is explicit:

<code>auth_state</code>	Frame has tag	Ground station has key	Result
<code>NOT_CONFIGURED</code>	no	no	accepted as before
<code>AUTHENTICATED</code>	yes	yes, matches	accepted
<code>UNAUTHENTICATED</code>	no	yes	SUSPECT; INVALID when telemetry.require_authenticated_frames is true
<code>UNVERIFIABLE</code>	yes	no	SUSPECT

auth_state	Frame has tag	Ground station has key	Result
			T — cannot be checked
INVALID_TAG	yes	yes, mismatch	AuthTag Error; emitted as a corrupted packet with no measurements

The tag provides integrity and origin authentication only — the payload is **not** encrypted (FLAG_ENCRYPTED remains reserved). Replay is handled by sequence tracking: a sequence far behind the newest one is flagged replay = True and counted.

Stream parsing

StreamParser.parse_stream(data: bytes) -> List[ParsedPacket] is resync-safe:

- It scans for the MAGIC sync word, so partial or noisy streams recover at the next valid frame boundary rather than desynchronising permanently.
- Each ParsedPacket exposes .header (dict), .payload (bytes), .crc_valid, .duplicate, .out_of_order, .replay, .auth_state, .rejected, and a .sequence property. StreamParser.counters() exposes all diagnostic counters.
- Duplicate and out-of-order frames are **flagged, not silently dropped**, so the quality accounting downstream stays honest.

Forward Error Correction (optional)

When FLAG_FEC_PRESENT is set and the reedsolo package is installed, the pipeline's FEC stage attempts Reed-Solomon recovery. If reedsolo is **not** installed, FEC is disabled explicitly and reported — corrupted frames are marked CORRUPTED rather than silently passed through as if valid.

Decoding to measurements

decode_payload_to_frame(pkt) converts a ParsedPacket into a TelemetryFrame of fully-provenanced Measurement objects. Each measurement carries its value, unit, timestamp, sensor id, QualityState, DataSource, and uncertainty — never a bare number.

Payload limits (hostile input never raises): UTF-8 JSON object, ≤ 16 KiB, ≤ 64 fields, nesting depth ≤ 3 , strings ≤ 64 chars. Non-finite or non-numeric values become INVALID; a declared source may only downgrade provenance to SIMULATED; implausible timestamps are replaced by receive time and flagged timestamp_replaced.